

Does causal probing resist spurious features better than diagnostic probing?

A finite-complexity analysis

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<https://github.com/tpimentelms/diagnostic-vs-causal-probing>

The question

Two recent results put diagnostic and causal probing in the same uncomfortable place. (Throughout this report, we refer to distributed alignment search, an important method from mechanistic interpretability, as causal probing—making it a direct analogous to diagnostic probing.) A sufficiently powerful diagnostic probe always recovers a target property from a model’s representations (Pimentel et al. 2020); and a sufficiently complex alignment map makes *any* algorithm a distributed abstraction of *any* network (Sutter et al. 2025). Both are statements about a limit: at infinite complexity either method can fit any target—random ones included—so the methods’ reported scores say nothing about the network. Read at face value, these results say that causal probing is no better than diagnostic probing: it leaves us exactly where diagnostic probing did.

But this is a claim about a limit at which we never work. At finite complexity and finite data neither method can perfectly fit arbitrarily complex datasets, and the extra constraints imposed in a causal probe—it must act through the frozen network rather than emit a label directly—might make it harder to fit *spurious* patterns than a diagnostic probe of comparable power. If so, causality may still help in practice even though both methods fail in the limit. This note asks whether it does.

A shared measure of spuriousness

In this report, we measure how readily a method fits a structure that is not there, randomly relabelling a dataset, and seeing how well the method still fits it. The expected best fit a hypothesis class achieves on such random targets is, up to scaling, its empirical Rademacher complexity (Bartlett and Mendelson 2002). We use it, in accuracy space, as a *spurious-fitting capacity* that applies unchanged to both methods: we fit random labels for a diagnostic probe, and a randomly relabelled counterfactual

target for causal probing (DAS; Geiger et al. 2024). Notably, in the limit of infinitely complex probes both methods fail; however, they may approach this failure mode at different rates. In this case, the method that fits random labels less readily is the safer to use for model analyses.

Setup

We use the hierarchical equality task of Geiger et al. (2024): the input is two pairs of vectors and the label is whether the pairs are *both* equal or *both* unequal, $O = (W = X) = (Y = Z)$. We train a small MLP to solve it and study the intermediate concept $WX = (W = X)$. The diagnostic probe is a linear classifier on the MLP’s first-layer activations; causal probing is implemented as linear DAS—a learned rotation of those activations with an interchange intervention on the WX subspace. We sweep the number of training examples n and record each method’s *train-set fit* on the correct labels (signal) and on a fixed random relabelling (spurious capacity).

Findings

Both methods recover the concept on the correct labels: the probe decodes WX at ≈ 0.98 , and linear DAS reaches ≈ 0.999 interchange-intervention accuracy. So the feature is both *decodable* and *interveneable*.

The comparison that matters is on the random labels, however (Figure 1, dashed). Both methods memorise the smallest dataset ($n = 30$) and both decay toward chance as n grows. In between—where capacity, not data, is the binding constraint—DAS fits the random labels *less* readily than the probe at most sizes, most clearly at $n = 90$ (probe ≈ 0.86 , DAS ≈ 0.66). Taken at face value, DAS reaches the same signal while fitting noise less readily: its spurious-fitting capacity is the smaller of the two. That is the direction the finite-complexity hypothesis predicts—causal probing as the safer tool—even though, in the limit, both are equally vacuous.

We report this as a preliminary result, however, not settling it. The presented curves below are from a single seed and noisy—they cross once (at $n = 300$), and DAS’s correct-label fit wobbles at the smallest n —so the gap, which rests largely on the $n = 90$ point, needs averaging over seeds before it is leaned on. Further, exploring how this curve changes with the complexity of the probe architecture (e.g., when using MLPs for our diagnostic probes, or Reversible Networks as our causal probes) would also be interesting.

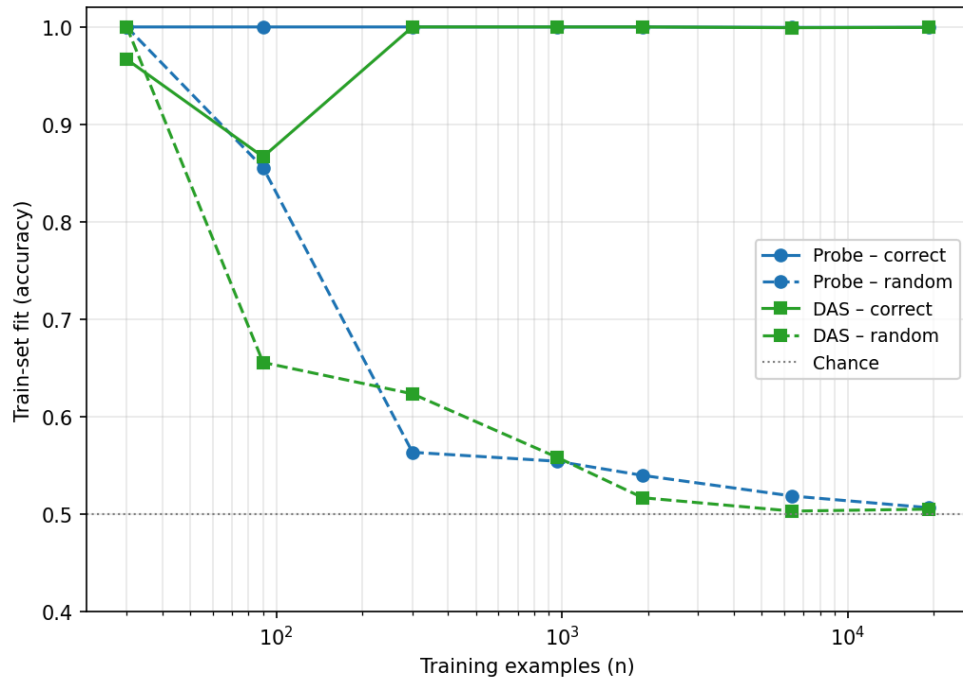


Figure 1. Train-set fit vs. number of training examples n , for the diagnostic probe and linear DAS, on correct labels (signal, solid) and a fixed random relabelling (spurious capacity, dashed).

References

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